POWER_FEATURES_PORTING_PLAN

POWER FEATURES PORTING PLAN — CLI-Agent Power Features → HelixCode

Field	Value
Revision	2
Created	2026-06-14
Last modified	2026-06-14
Status	DRAFT — research + plan only, NO code (operator decision pending). **Rev 2: Phase-1 status reconciled against HEAD `e7266263` — T1.1/T1.2/T1.3 found already DONE; T1.4/T1.5/T1.6 PARTIAL; stale paths corrected (`internal/autonomy`→`internal/workflow/autonomy`, `internal/planner`→`internal/workflow/planmode`).**
Authority	§11.4.70 subagent-driven, §11.4.74 catalogue-first, §11.4.6 no-guessing, §11.4.50/§11.4.98 anti-bluff
Method	Read-only research of `cli_agents/<agent>/` READMEs+docs; cross-checked against `helix_code/internal/*` + `helix_code/applications/*`
Scope	Map distinctive power features of every in-repo CLI agent for porting into HelixCode

---

Table of Contents

---

1. Executive Summary

- **Agents enumerated under `cli_agents/`:** 51 directories. Of these, **~38 are coding/agent tools** worth mining; the remainder are MCP servers (`git-mcp`, `postgres-mcp`, `conduit`, `noi`, `snow-cli`, `x-cmd`, `cli-agent`, `aiagent`, `superset`, `ui-ux-pro-max`, `fauxpilot`, `deepseek-cli*`, `xela-cli`, `codex-skills`) or a skills/plugin marketplace (`bridle` = "Tons of Skills") rather than full agents.
- **Distinctive power features mapped: ~140** across the 24 agents researched in depth (aider, cline, plandex, crush, codex, gemini-cli, qwen-code, copilot-cli, amazon-q, mistral-code, codename-goose, gptme, open-interpreter, forge, gpt-engineer, swe-agent, shai, nanocoder, claude-squad, claude-code, vtcode, warp, get-shit-done, spec-kit, junie, aichat, codai, taskweaver, octogen, zeroshot, multiagent-coding, agent-deck). After de-duplicating equivalent features across agents, they collapse into **~62 distinct capabilities**.
- **HelixCode is already mature.** It HAS: agentic tool loop (`internal/agent`), read-only tools + browser suite (`internal/tools`), MCP client+lifecycle (`internal/mcp`), plugins (`internal/plugins`), skills (via plugins + crush/vtcode-style `SKILL.md` precedent), LSP-equipped context, ensemble + verifier (`internal/verifier`), repomap (`internal/repomap`, tree-sitter), sessions + history-condense (`internal/session`), hooks (`internal/hooks`), sandbox + worktrees (`internal/worker`, `internal/session/manager_worktree`), memory + cognee (`internal/memory`), voice-to-code (`internal/voice`, the aider P2-F27 port), clarification (`internal/clarification`), task checkpoints (`internal/task/checkpoint.go`), roocode delegation (`internal/roocode`), kilocode callgraph/impact (`internal/kilocode`), planner/plantree (`internal/planner`, `internal/plantree`), approval/approvalwire, autocommit, git_status, multi-platform apps (desktop/terminal-ui/ios/android/aurora/harmony).
- **Net result:** Most "power features" are **HAVE** or **PARTIAL** — the real porting work is in **autonomy presets, cumulative-diff sandbox review, conversation branching, messaging connectors, daemon/server-shared-session mode, spec-driven workflow surface, and tangent/rewind UX**. Few features are genuinely **MISSING/large**.
- **Phasing:** 8 phases (0–7), ordered high-value-low-effort first, ending in niche features. Every feature ships with the operator's full anti-bluff bar (§7).

---

2. Method & Evidence Discipline

Per §11.4.6 (no-guessing) and §11.4.74 (catalogue-first), every feature row cites the source-agent file where the capability is textually evidenced. HelixCode status was cross-checked by reading helix_code/internal/<pkg>/doc.go, function signatures, and the existing port plan docs/plans/HXC-031-codex-cline-port.md. No source was modified. Where the in-repo doc only links out to a vendor website (codex docs/*.md stubs, mistral-code's inherited gemini docs), that is noted so we do not over-claim a feature exists in-tree.

Status legend: HAVE (shipped in HelixCode) · PARTIAL (substrate exists, surface/UX gap) · MISSING. **Effort legend:** S (small, ≤~300-line unit) · M (medium, 1–3 units) · L (large, multi-component).

---

3. What HelixCode Already Has (baseline)

Verified by reading the inner module (helix_code/internal, helix_code/applications):

Capability	HelixCode location	Evidence
Multi-agent orchestration + agent types	internal/agent	agent/doc.go (Coordinator, WorkflowExecutor, planning/coding/ testing/debug/review agents)
Repo-map (tree-sitter, ranked, incremental)	internal/repomap	repomap/file_ranker.go, cache.go, incremental_p2t06_test.go
MCP client + lifecycle + chaos-tested	internal/mcp	mcp/lifecycle.go, config.go, mcp_chaos_test.go
Plugins (loader, activation, registry, exec)	internal/plugins	plugins/loader.go, activation.go, registry.go
Ensemble + verifier (LLMsVerifier single source)	internal/verifier, internal/llm	verifier/client.go, adapter.go; ensemble commits in git log
Sessions + history compaction/condense	internal/session	session/condense.go (CompactIfNeeded, ShouldCompact), manager_worktree.go
Hooks (lifecycle blockers/executor)	internal/hooks	hooks/hook.go, blockers.go, executor.go
		worker/isolation_test.go, session/manager_worktree.go

Capability	HelixCode location	Evidence
Sandbox + worktree isolation	`internal/worker`, `internal/session`	
Memory + cognee + project memory	`internal/memory`, `internal/projectmemory`	`memory/manager.go`, `cognee_integration.go`, `projectmemory` `loader.go`
Voice-to-code (aider port P2-F27)	`internal/voice`	`voice/doc.go` (Whisper API + whisper.cpp fallback)
Clarification (LLM-generated questions)	`internal/clarification`	`clarification/engine.go` (`DetectAmbiguity`, `Resolve`)
Task checkpoints (DB-backed)	`internal/task`	`task/checkpoint.go` (`CreateCheckpoint/GetLatestCheckpoint`/restore)
Auto-commit + secret-filter + summariser	`internal/autocommit`	`autocommit/committer.go`, `secret_filter.go`, `summariser.g`
Roo-code delegation, Kilo-code callgraph/impact	`internal/roocode`, `internal/kilocode`	`roocode/delegator.go`, `kilocode/callgraph.go`, `impact.go`
Planner / plan-tree (compact, operations)	`internal/planner`, `internal/plantree`	`planner/executor.go`, `plantree/operations.go`
Approval engine + yes/no prompter	`internal/approval`, `internal/approvalwire`	`approval/manager.go`, `approvalwire/yesno_prompter.go`
Browser tools (chromedp, click/type/screenshot)	`internal/tools`	`tools/browser_*_v2.go`
Quality gate + scorer	`internal/quality`	`quality/gate.go`, `scorer.go`
Multi-platform apps + TUI	`applications/*`	`applications/{desktop,terminal_ui,ios,android,aurora_os,harmony_o`
Multimodal (in-flight per HXC-031)	`internal/llm/vision`, HXC-031 plan	`docs/plans/HXC-031-codex-cline-port.md`

This baseline means the plan is **mostly closing UX/surface gaps over existing substrate**, not building from zero.

---

4. Agents Surveyed

Full coding agents mined (24): aider, cline, plandex, crush, codex, gemini-cli, qwen-code, copilot-cli, amazon-q, mistral-code, codename-goose, gptme, open-interpreter, forge, gpt-engineer, swe-agent, shai, nanocoder, claude-code, vtcode, aichat, codai, taskweaver, octogen.

Orchestration / multi-agent managers (6): claude-squad, agent-deck, zeroshot, multiagent-coding, warp, get-shit-done.

Workflow / spec toolkits (2): spec-kit, junie.

Not full agents (noted, low-priority mining): `bridle` (skills/plugin marketplace), `git-mcp/postgres-mcp/conduit/noi/snow-cli/x-cmd/cli-agent/aiagent/superset/ui-ux-pro-max/fauxpilot/deepseek-cli*/xela-cli/codex-skills` (MCP servers / dashboards / stubs).

Confirmed ABSENT from `cli_agents/`: `openhands` (operator example) is NOT present — do not plan against it.

---

5. Master Feature Matrix

`feature` × `representative source-agent(s)` × `HelixCode status` × `effort` × `priority` (1=highest). Evidence column cites the canonical source file.

#	Feature	Source agent(s)	Evidence (cli_agents/…)	HelixCode s
F01	Repo-map (tree-sitter project map)	aider, plandex, codai	`aider/aider/website/docs/repomap.md`; `plandex/README.md`; `codai/README.md`	HAVE (inte
F02	Git auto-commit w/ generated message	aider, plandex, forge	`aider/README.md`; `plandex/docs/.../version-control.md`; `forge/README.md` (`:commit`)	HAVE (inte
F03	Architect/editor dual-model mode	aider	`aider/aider/website/docs/usage/modes.md`	PARTIAL (e no explicit a handoff)
F04	Chat modes (code/ask/architect/plan)	aider, plandex, gemini-cli, forge	`aider/.../modes.md`; `plandex/.../repl.md`; `gemini-cli/docs/cli/plan-mode.md`; `forge` (sage/muse)	PARTIAL (p tools; no firs
F05	Voice-to-code	aider	`aider/.../voice.md`	HAVE (inte

#	Feature	Source agent(s)	Evidence (cli_agents/…)	HelixCode s
F06	Lint & test on edit (auto-fix loop)	aider, cline, swe-agent	`aider/.../lint-test.md`; `cline/README.md`; `swe-agent/docs/background/aci.md`	PARTIAL (c wired as per-
F07	`/undo` + `/diff` (git-aware)	aider, forge	`aider/.../commands.md`	HAVE (DON `cli/ main.go:21` `internal/a` `git.go:85,`
F08	Images & web pages in chat (`/web`,`/paste`)	aider, gpt-engineer	`aider/.../images-urls.md`; `gpt-engineer/README.md` (`--image_directory`)	PARTIAL (I in-flight; `/w` tools)
F09	Watch-files / IDE-comment mode	aider	`aider/.../watch.md`	MISSING
F10	Copy/paste-to-web-chat (`/copy-context`)	aider	`aider/.../copypaste.md`	MISSING
F11	Plan/Act dual mode	cline, gemini-cli	`cline/README.md`; `gemini-cli/docs/cli/plan-mode.md`	HAVE (DON `internal/w` `mode_contr` wired `cmd/c` `main.go:77`
F12	Checkpoints (workspace snapshot + restore/undo)	cline, gemini-cli, amazon-q, nanocoder	`cline/README.md`; `gemini-cli/docs/checkpointing.md`; `amazon-q/docs/experiments.md`; `nanocoder/docs/features/index.md`	PARTIAL (t not workspa restore)
F13	MCP marketplace + on-the-fly tool creation	cline, codename-goose	`cline/README.md`; goose extension system	PARTIAL (M marketplace
F14	`.clinerules/GEMINI.md/AGENTS.md` context files	cline, gemini-cli, codex, forge, shai, mistral-code	`cline/.clinerules/*`; `gemini-cli/GEMINI.md`; `codex/AGENTS.md`; `shai/README.md` (SHAI.md)	HAVE (inte `projectmem`
F15	Plugin SDK (programmatic tools + hooks)	cline, gptme	`cline/README.md` (`@cline/sdk`); `gptme/README.md`	HAVE (inte
F16	Multi-agent teams (coordinator → specialists)	cline, codename-goose, multiagent-coding, zeroshot	`cline/README.md`; goose GooseTeam; `multiagent-coding/README.md`	PARTIAL (a roocode dele team state)
F17	Scheduled/cron agents	cline, nanocoder	`cline/README.md`; `nanocoder` daemon (croner)	MISSING

#	Feature	Source agent(s)	Evidence (cli_agents/…)	HelixCode s
F18	Messaging connectors (Slack/ Telegram/Discord/…)	cline	`cline/README.md`	MISSING (I precedent ex
F19	Headless / non-interactive (JSON/ stream-json out)	cline, gemini-cli, qwen-code, codex, shai, vtcode, gpt-engineer	`cline/README.md`; `gemini-cli/docs/cli/headless.md`; `codex/docs/exec.md`; `shai/README.md`; `vtcode/README.md`	PARTIAL (c structured JS output)
F20	Background long-process monitoring	cline, plandex	`cline/README.md`; `plandex/.../execution-and-debugging.md`	PARTIAL (i `backgrounc`
F21	Kanban / parallel task board	cline, agent-deck	`cline/README.md`; `agent-deck/llms.txt`	MISSING
F22	Cumulative diff-review sandbox (stage-before-apply)	plandex, claude-squad, forge	`plandex/.../reviewing-changes.md`; `claude-squad/README.md`; `forge/README.md` (`--sandbox`)	PARTIAL (v cumulative-c reject UX)
F23	Configurable autonomy levels (preset tiers)	plandex, nanocoder, copilot-cli	`plandex/.../autonomy.md`; `nanocoder` modes; `copilot-cli` Autopilot	HAVE (DON `internal/w` `modes.go:1` presets)
F24	Automated command/build/test debugging	plandex, shai, gptme	`plandex/.../execution-and-debugging.md`; `shai/README.md` (shell hook); `gptme`	PARTIAL (v auto-debug l
F25	Conversation branches / fork	plandex, forge, claude-squad, agent-deck	`plandex/.../branches.md`; `forge/README.md`; `agent-deck/llms.txt`	MISSING (s branch/fork)
F26	Big-context management (load files/dirs/urls; smart context)	plandex, gemini-cli, nanocoder	`plandex/.../context-management.md`; `gemini-cli/README.md`; `nanocoder` File Explorer	HAVE/PART repomap; `@-` UX partial)
F27	Model packs / mid-session model switch	plandex, crush, copilot-cli, junie, qwen-code	`plandex` model packs; `crush/README.md`; `copilot-cli /model`; `qwen-code /model`	HAVE (LLM `llm/aliase`
F28	LSP-enhanced context	crush, copilot-cli, nanocoder	`crush/README.md`; `copilot-cli/README.md`; `nanocoder` (`source/lsp/`)	HAVE (LSP
F29	Permission allowlist + `--yolo`/ auto-approve	crush, vtcode, open-interpreter,	`crush/README.md`; `vtcode/README.md`; `open-interpreter/README.md` (`-y`)	HAVE (`inte`

#	Feature	Source agent(s)	Evidence (cli_agents/…)	HelixCode s
		claude-squad, qwen-code		
F30	Provider auto-update catalogue	crush (Catwalk)	`crush/README.md`	HAVE (LLM source, CON
F31	Shared workspace across clients (server backend)	crush, qwen-code (daemon), shai (serve)	`crush/README.md`; `qwen-code/README.md`; `shai/README.md`	MISSING (s shared-sessi
F32	Self-config skill ("configure yourself")	crush	`crush/README.md` (crush-config skill)	PARTIAL (c LLM-driven
F33	OS sandbox for tool exec (Seatbelt/ Landlock/bwrap)	codex, gemini-cli, octogen, taskweaver	`codex/AGENTS.md`; `codex/docs/sandbox.md`; `octogen/README.md`; `taskweaver` container	PARTIAL (v infraboot; O landlock not
F34	Exec-policy / command-approval engine	codex, amazon-q, vtcode	`codex/docs/execpolicy.md`; `amazon-q/docs/built-in-tools.md`	HAVE (`inte` selector)
F35	ACP mode (Agent Client Protocol over stdio)	gemini-cli, gptme, qwen-code	`gemini-cli/docs/cli/acp-mode.md`; `gptme/README.md`; `qwen-code` daemon (ACP)	MISSING
F36	`/rewind` (conversation + file-state rewind)	gemini-cli	`gemini-cli/docs/cli/rewind.md`	PARTIAL (c UX)
F37	Model routing / auto-fallback on quota	gemini-cli, crush	`gemini-cli/docs/cli/model-routing.md`	HAVE (`inte` `auto_llm_m` mode)
F38	Model steering (mid-task course-correct)	gemini-cli	`gemini-cli/docs/cli/model-steering.md`	MISSING
F39	Git worktrees per session	gemini-cli, claude-squad, forge, zeroshot, agent-deck	`gemini-cli/docs/cli/git-worktrees.md`; `claude-squad/README.md`	HAVE (`inte` `manager_wo`
F40	Agent Skills (`SKILL.md` open standard)	gemini-cli, crush, vtcode, forge, nanocoder, codex, qwen-code	`gemini-cli/docs/cli/skills.md`; `crush/README.md`; `vtcode/README.md`; `forge/README.md`	PARTIAL (2 subsystem sl `commands/m` 2-tier projec `:253–254`; M `SKILL.md` fi
F41	`ask_user` / TODO / tracker tools	gemini-cli, amazon-q, nanocoder	`gemini-cli/docs/tools/{ask-user,todos,tracker}.md`; `amazon-q/docs/experiments.md`	PARTIAL (2 DONE `inte` `askuser/as` wired `cmd/c` `main.go:70`

#	Feature	Source agent(s)	Evidence (cli_agents/…)	HelixCode s
				background-tracker)
F42	Knowledge base (persistent semantic index)	amazon-q, aichat (RAG), gptme (RAG), forge (semantic search)	`amazon-q/docs/knowledge-management.md`; `aichat/README.md`; `forge/README.md` (`:sync`)	HAVE (inte + RAG)
F43	Tangent mode (side-topic checkpoint)	amazon-q	`amazon-q/docs/tangent-mode.md`	MISSING
F44	Introspect tool (self-aware about own features)	amazon-q	`amazon-q/docs/introspect-tool.md`	MISSING
F45	Delegate (async background sub-sessions)	amazon-q, gptme (subagent), nanocoder, vtcode	`amazon-q/docs/experiments.md`; `gptme/README.md`; `vtcode/README.md`	PARTIAL (r workflow ba
F46	Recipes (reusable parametrized workflows)	codename-goose, forge (skills)	goose recipes; `forge/README.md`	PARTIAL (p parametrized
F47	Lessons system (auto-injected contextual guidance)	gptme	`gptme/README.md`	MISSING
F48	Computer use (GUI/desktop control)	gptme, open-interpreter, warp (Oz)	`gptme/README.md`; `open-interpreter/README.md`	PARTIAL (b desktop cont
F49	Local code-exec REPL (NL → run code)	open-interpreter, gptme, octogen, taskweaver	`open-interpreter/README.md`; `octogen/README.md`	HAVE (com BLUFF-003
F50	Profiles (YAML behavior presets)	open-interpreter, codename-goose, claude-squad	`open-interpreter/README.md`; goose profiles	PARTIAL (c named beha
F51	Whole-project generation from a prompt	gpt-engineer	`gpt-engineer/README.md`	PARTIAL (a `gpte <dir>` mode)
F52	Benchmark harness (SWE-bench/TerminalBench)	gpt-engineer, swe-agent, codename-goose,	`gpt-engineer/README.md`; `swe-agent/README.md`; goose `goose-bench`	MISSING (h agent-bench

#	Feature	Source agent(s)	Evidence (cli_agents/…)	HelixCode s
		multiagent-coding		
F53	Agent-Computer Interface tools (windowed editor/viewer, succinct search)	swe-agent	`swe-agent/docs/background/aci.md`	PARTIAL (e windowed-v
F54	Shell assistant (failed-command auto-fix hook)	shai, aichat, plandex	`shai/README.md`; `aichat/README.md`	MISSING
F55	Chainable `--trace` (pipe conversation between runs)	shai	`shai/README.md`	MISSING
F56	HTTP server (OpenAI-compatible endpoints)	shai, aichat, gptme, open-interpreter, octogen	`shai/README.md`; `aichat/README.md`; `gptme/README.md`	PARTIAL (i OpenAI-con unconfirmed
F57	Context compression `/compact` + `/usage`	nanocoder, qwen-code, forge	`nanocoder/docs/features/index.md`; `qwen-code/README.md`	HAVE (inte `condense.g`
F58	Session auto-save + `/resume`	nanocoder, codex, vtcode	`nanocoder/docs/features/index.md`; `vtcode/README.md`	HAVE (inte
F59	Event triggers (file-watch + cron → skill)	nanocoder	`nanocoder/CLAUDE.md` (daemon)	MISSING
F60	VS Code / IDE extension bridge	nanocoder, codex, qwen-code, junie	`nanocoder/docs/features/vscode-extension.md`; `codex/README.md`	MISSING (c for app team
F61	Spec-driven workflow (specify→plan→tasks→implement)	spec-kit, get-shit-done	`spec-kit/README.md`; `get-shit-done/docs/FEATURES.md`	PARTIAL (p spec-comma
F62	QA / regression gates in workflow	get-shit-done	`get-shit-done/docs/FEATURES.md`	HAVE (qual helix_qa)
F63	Cross-AI peer review (multi-runtime review)	get-shit-done, multiagent-coding	`get-shit-done/docs/FEATURES.md`	PARTIAL (e cross-runtim
F64	Token/usage tracking per request + context-%	codai, amazon-q, nanocoder	`codai/README.md`; `amazon-q/docs/experiments.md`	HAVE/PART in-prompt %
F65	Stateful code-first execution (DataFrames persist)	taskweaver, octogen	`taskweaver/README.md`; `octogen/README.md`	PARTIAL (s analytics-sta
F66	MCP socket pool (share MCP procs across sessions)	agent-deck	`agent-deck/llms.txt`	MISSING
F67	Global conversation search (fuzzy + regex)	agent-deck	`agent-deck/llms.txt`	MISSING

#	Feature	Source agent(s)	Evidence (cli_agents/…)	HelixCode s
F68	Conversation export (`:dump` JSON/ HTML)	forge, aichat	`forge/README.md`	MISSING
F69	Restricted/secure shell mode (path/ symlink guards)	forge, vtcode	`forge/README.md`; `vtcode/README.md` (layered security)	PARTIAL (a pkg)
F70	ChatGPT/account-plan auth (non-API-key sign-in)	codex, junie, copilot-cli	`codex/docs/authentication.md`; `copilot-cli/README.md`	OUT-OF-SC specific; ver

---

6. Top-10 Highest-Value Missing Features

Ranked by (user impact × low duplication-with-existing × evidence strength). “Missing” or “Partial-surface” only — excludes things HelixCode already HAS.

1. **F22 — Cumulative diff-review sandbox (stage-before-apply, apply/reject hotkeys)** — plandex/forge/claude-squad. The single biggest trust/UX lever; HelixCode has worktrees but no cumulative-diff staging+review surface. **L.**
2. ~~**F23 — Configurable autonomy levels (named preset tiers None →Full)**~~ — **DONE 2026-06-14** (`internal/workflow/autonomy/modes.go:12–31`, 5 named presets). No longer a missing feature; kept here only for ranking history.
3. **F12 — Workspace checkpoints (file snapshot + restore/undo)** — cline/gemini/amazon-q/nanocoder. HelixCode checkpoints are task-DB rows, not workspace-file snapshots; this is the “oops, revert everything” safety net. **M.** (Phase 2 — not re-verified this 2026-06-14 pass.)
4. **F11/F04 — Plan/Act + first-class chat modes (code/ask/architect/plan)** — cline/aider/gemini/forge. **F11 Plan/Act is DONE 2026-06-14** (`internal/workflow/planmode/mode_controller.go:13–14`, wired `cmd/cli/main.go:773–779`); F04’s broader code/ask/architect mode menu remains the open surface. **M.**
5. **F61 — Spec-driven workflow surface (specify→clarify→plan→tasks→implement + constitution)** — spec-kit/get-shit-done. Aligns directly with HelixCode’s constitution discipline. **L.**
6. **F25 — Conversation branches / fork** — plandex/forge/agent-deck. Explore multiple approaches/models from a shared point. **M.**
7. **F36/F43 — `/rewind` + Tangent mode** — gemini-cli/amazon-q. Side-explore and time-travel without losing place. **M.**
8. **F19/F56 — Robust headless (JSON/stream-json) + OpenAI-compatible server** — cline/gemini/shai/aichat. Unlocks CI/CD + programmatic embedding. **M.**
9. **F33 — OS-level tool-exec sandbox (Seatbelt/Landlock/bwrap)** — codex/gemini. Hard isolation beyond worktrees; safety-critical for full-auto. **L.**
10. **F18 — Messaging connectors (Slack/Telegram/Discord/Linear)** — cline. Org already has the Herald precedent to extend (§11.4.74 catalogue-first), making this lower-risk than greenfield. **L.**

Runners-up (just outside top-10): F35 ACP mode, F45 async delegate, F59 event triggers, F54 shell-assistant auto-fix.

---

7. Per-Feature Acceptance Bar (anti-bluff)

Per the operator's bar and §11.4.27 / §11.4.69 / §11.4.83 / §11.4.98 / §11.4.107 / §11.4.135, **every** ported feature is DONE only when it ships ALL of:

1. **Real implementation** — no simulation/placeholder (§Rule 2, BLUFF-001/002/003); touches real files/processes/LLM calls.
2. **A triggering prompt** — the exact natural-language prompt that exercises the feature end-to-end (§11.4.105 intent-recognition compatible).
3. **A recorded video/transcript demo** under `docs/qa/<run-id>/` (§11.4.83) — full bidirectional thread + captured evidence.
4. **Tests** — unit + integration + (where applicable) stress + chaos (§11.4.85), each anti-bluff with a paired §1.1 mutation; integration/E2E hit real infra (§11.4.50(A)).
5. **A Challenge** in `challenges/` exercising the complete user workflow (§Rule 6/7).
6. **A HelixQA bank entry** executed in an autonomous session with captured wire evidence (§11.4.27 (B)).
7. **A standing regression guard** registered on close (§11.4.135) with RED-on-broken-artifact polarity switch (§11.4.115).
8. **Docs in sync** — feature manual/guide + tracker row with `Catalogue-Check:` field (§11.4.74).

---

8. Phases → Tasks → Sub-Tasks

Each task names concrete file targets (under `helix_code/internal/…` or `helix_code/cmd/cli/…`) and acceptance criteria. Tasks are sized for §11.4.70 subagent-driven, ≤~300-line units. Every sub-task inherits the §7 acceptance bar.

Phase 0 — Foundations & Audit

> Goal: replace assumptions with FACTS before any port. No new features.

- **T0.1 — Capability-confirmation audit.** Verify the PARTIAL rows by reading code, not guessing (§11.4.6).
  - ST0.1.1 — Confirm headless JSON/stream-json output state in `cmd/cli/main.go` + `internal/render`. Acceptance: documented yes/no with file:line.
  - ST0.1.2 — Confirm `SKILL.md` precedence vs plugins in `internal/plugins` (F40). Acceptance: precedence table captured.
  - ST0.1.3 — Confirm OS-sandbox depth in `internal/worker/isolation` + `internal/infraboot` (F33). Acceptance: "worktree-only" vs "seatbelt/landlock" determination.
  - ST0.1.4 — Confirm OpenAI-compat surface in `internal/server` (F56). Acceptance: endpoint inventory.
- **T0.2 — Catalogue-check sweep (§11.4.74).** For each planned feature, record `reuse|extend|no-match <org/repo@sha>` (e.g. F18 → extend Herald; F42 → reuse cognee). Acceptance: every Phase 1–6 feature has a `Catalogue-Check:` line.

- **T0.3 — Tracker rows.** Open one Feature item per planned feature in the workable-items DB (§11.4.93) with `created_by`/`assigned_to` (§11.4.104). Acceptance: DB⍰MD in sync (`workable-items validate`).

Phase 1 — High-Value / Low-Effort (already-partial)

> Wrap existing substrate into discoverable surfaces. Mostly S/M.

> ⚠ **Reconciled 2026-06-14 against HEAD `e7266263`** (read-only grep/read of the real inner tree, file:line cited per §11.4.6). Phase-1 was STALE: most items the plan filed as "to do / partial" are already SHIPPED — several under different package paths than the plan named, and T1.1's CLI wiring (plan said "pending") is in fact live. Per-item ground-truth status table is below; corrected paths are inlined. Future-phase content (Phases 2–7) was NOT re-verified this pass.

Item	Status (2026-06-14)	Evidence (file:line)	What was wrong in the original plan
T1.1 `/undo`+`/diff`	**DONE**	handlers `helix_code/cmd/cli/main.go:2118–2133` (dispatch) + `:2326–2371` (`handleDiff`/`handleUndo`); real git via `helix_code/internal/autocommit/git.go:85 DiffUnstaged`, `:136 RevertLastCommit`, `:151 DiffSinceRef`	Plan said substrate-only "no `/undo`/`/diff` UX" and prior audit said "CLI wiring still pending" — both WRONG; both commands are fully wired and use real git output.
T1.2 Autonomy presets	**DONE**	`helix_code/internal/workflow/autonomy/modes.go:12–31` (5 named modes None/Basic/BasicPlus/SemiAuto/FullAuto), `:137 Level()` 1-5, capability table `:51–122`	Plan named the package `internal/autonomy`; it actually lives at `internal/workflow/autonomy/`.
T1.3 Plan/Act modes	**DONE**	`helix_code/internal/workflow/planmode/mode_controller.go:13–14` (`ModePlan`/`ModeAct`), transition table `:118–121`; wired `helix_code/cmd/cli/main.go:773–779` (`NewModeController`, `NewToolGate`, enter/exit plan-mode tools)	Plan named `internal/planner` for the mode controller; the first-class Plan/Act controller is `internal/workflow/planmode/` and is already CLI-wired.
T1.4 TODO/ask_user surface	**PARTIAL**	`ask_user` DONE: `helix_code/internal/tools/askuser/ask_user_tool.go:47–61`, wired `helix_code/cmd/cli/main.go:707–714`. `/tasks` exists (`helix_code/internal/commands/tasks_command.go:29`, wired	Plan implied the TODO surface bridges `internal/plantree`+`internal/clarification`; the shipped `/tasks` is a background-job tracker. The plantree slash command is `plantree` (`internal/commands/`

Item	Status (2026-06-14)	Evidence (file:line)	What was wrong in the original plan
		`main.go:769`) but wraps `workflow.BackgroundManager` (background-job list), NOT the plantree/clarification TODO-tracker this item describes.	`plan_tree_command.go:20`), not a TODO `/tasks`.
T1.5 Token + context-%	**PARTIAL**	per-request token count DONE: `helix_code/cmd/cli/main.go:1727–1740` (`printGenerationStats`, real `resp.Usage.PromptTokens/CompletionTokens`, anti-bluff “don’t fabricate”). Context-% indicator **ABSENT** — no production hit for any context-window-percent render in `internal/render` or `cmd/cli`.	Plan filed F64 as “HAVE/PARTIAL”; the token half is DONE, the context-% half is genuinely ABSENT (only the token counter ships).
T1.6 Skill `SKILL.md` precedence	**PARTIAL**	markdown-skills subsystem ships: `helix_code/internal/commands/markdown_skills.go` (`SkillLoader :217`, precedence `:253–254` “user first, then project; project overrides user”), wired `helix_code/cmd/cli/main.go:991–1010` (`.helix/skills/` + user-config dir).	NOT absent. Gaps vs the AC: (a) filename is any `*.md` (`markdown_skills.go:267`), not the `SKILL.md` open-standard filename; (b) precedence is **2-tier** (project>user) — the **built-in/bundled tier is MISSING** (no `//go:embed`, no default skills). So “repo/user/built-in precedence” is only 2 of 3 tiers.

- **T1.1 — `/undo` + `/diff` (F07, S). ✅ DONE (2026-06-14).** Shipped: handlers at `cmd/cli/main.go:2118–2133` + `:2326–2371`, bridged to real git in `internal/autocommit/git.go` (`DiffUnstaged`:85, `RevertLastCommit`:136, `DiffSinceRef`:151). `/undo` does `git revert --no-edit HEAD` (force-push-free per §11.4.113); `/diff` prints verbatim `git diff [<ref>]`. Remaining (if AC tightening desired): the dedicated reproduction Challenge reverting a real edit. (NOTE: the prior audit’s “CLI wiring still pending” is corrected to DONE.)
- **T1.2 — Autonomy presets (F23, M). ✅ DONE (2026-06-14).** Shipped at `internal/workflow/autonomy/` (NOT `internal/autonomy` — path corrected). `modes.go:12–31` defines the five named presets (None/Basic/BasicPlus/SemiAuto/FullAuto), `Level()`:137 maps them to levels 1-5, and `GetCapabilities`:51-122 drives per-tier behaviour. AC residual: confirm the paired §1.1 mutation (flip a tier → test FAILs) is registered.
- **T1.3 — First-class chat/Plan-Act modes (F04, F11, M). ✅ DONE (2026-06-14).** Shipped at `internal/workflow/planmode/` (NOT `internal/planner` — path corrected). `mode_controller.go:13–14` (`ModePlan`/`ModeAct`), legal transitions `:118–121`; CLI-wired at `cmd/cli/main.go:773–779` via `NewModeController` + `NewToolGate` + enter/exit-plan-mode tools (tool availability gated by mode). AC residual: the “no writes in ask/plan mode” Challenge.
- **T1.4 — TODO/tracker tool surface (F41, S). ◑ PARTIAL (2026-06-14).** `ask_user` DONE (`internal/tools/askuser/ask_user_tool.go:47–61`,

wired `cmd/cli/main.go:707–714`). The plan's TODO-tracker (plantree/clarification → `/tasks`) is NOT what shipped: `/tasks` (`internal/commands/tasks_command.go:29`, wired `main.go:769`) lists **background jobs** via `workflow.BackgroundManager`, not a step-by-step TODO tracker. Remaining: wire `internal/plantree` (slash command currently named `plantree`, `internal/commands/plan_tree_command.go:20`) + `internal/clarification` into a user-visible task-tracker surface.
- **T1.5 — Token/usage + context-% indicator (F64, S). ◑ PARTIAL (2026-06-14).** Per-request token count DONE: `cmd/cli/main.go:1727–1740` (`printGenerationStats`) prints real `resp.Usage.PromptTokens/CompletionTokens` from the provider `usage` object (anti-bluff: refuses to fabricate when unpopulated, §11.4.141-compliant). Remaining: the **context-window-% indicator is ABSENT** — no production code renders a context-usage percentage; this is the genuine outstanding work for T1.5.
- **T1.6 — Skill `SKILL.md` precedence (F40, S, gated on ST0.1.2). ◑ PARTIAL (2026-06-14).** A markdown-skills subsystem already ships (`internal/commands/markdown_skills.go` + `skills_command.go` + `skills_watcher.go`, wired `cmd/cli/main.go:991–1010`, dirs `.helix/skills/` + user-config-dir). It implements **2-tier** precedence (project overrides user, `markdown_skills.go:253–254`). Remaining vs the AC: (a) the **built-in/bundled tier** is missing (no `//go:embed` defaults) → "repo/user/built-in" is only 2 of 3 tiers; (b) the loader keys on any `*.md` filename (`:267`), not the `SKILL.md` open-standard filename. Document/implement the third tier + filename convention.

Phase 2 — Session & Context Power Tools

- **T2.1 — Workspace checkpoints (F12, M).** Targets: extend `internal/session` + worktree to snapshot/restore working-tree files (not just task DB). AC: create→edit→restore round-trips real files; chaos test mid-write SIGKILL recovers cleanly.
- **T2.2 — Conversation branches/fork (F25, M).** Targets: `internal/session` branch primitive + `cmd/cli` `/branch`. AC: fork from a point, diverge, compare; both branches persist.
- **T2.3 — `/rewind` (F36, M).** Targets: compose T2.1 checkpoints + history rewind in `internal/session/condense.go`. AC: rewind reverts conversation + optionally files.
- **T2.4 — Tangent mode (F43, M).** Targets: `internal/session` tangent stack. AC: enter tangent, explore, return to exact prior state; `tangent tail` keeps last Q+A.
- **T2.5 — Big-context `@`-mention + file explorer (F26, M).** Targets: `internal/context` + repomap; `@file`/`@dir`/`@url` loaders + token estimates. AC: `@`-load injects real content with token budget shown.
- **T2.6 — Conversation export + global search (F68, F67, S each).** Targets: `internal/session` `:dump` JSON/HTML; fuzzy+regex search across sessions. AC: exported HTML opens; search finds a known phrase.

Phase 3 — Autonomy & Safety Surface

- **T3.1 — OS-level exec sandbox (F33, L, gated on ST0.1.3).** Targets: `internal/worker/internal/infraboot` — Seatbelt (macOS), Landlock/bwrap (Linux), per §11.4.81 cross-platform parity (per-OS branches + honest kernel-gap citation). AC: a write outside the sandbox is blocked on each OS branch; paired mutation strips a branch → gate FAILs.

- **T3.2 — Lint/test-on-edit auto-fix loop (F06, M).** Targets: `internal/editor` + `internal/quality` — block edit on syntax/lint failure, auto-retry. AC: a deliberately broken edit is blocked then fixed; SWE-agent-style linter-block evidenced.
- **T3.3 — Automated command/build/test debugging (F24, M).** Targets: `internal/verifier` + new debug loop on failing exec. AC: a failing build is auto-diagnosed + fixed with captured evidence.
- **T3.4 — Shell-assistant failed-command hook (F54, M).** Targets: `cmd/cli` shell hook + `internal/approval`. AC: a failed shell command yields an LLM-proposed fix (opt-in).
- **T3.5 — Restricted/secure shell mode (F69, M).** Targets: extend `internal/security` path/symlink/dangerous-command guards. AC: a path-escape command is blocked; audit log captured.

Phase 4 — Multi-Agent & Parallel Orchestration

- **T4.1 — Cumulative diff-review sandbox (F22, L).** Targets: new `internal/changeset` staging layer over worktrees + `cmd/cli` review menu (apply/reject per-file/hunk). AC: changes accumulate unapplied; `diff --ui` apply/reject lands only accepted hunks; reject leaves tree clean.
- **T4.2 — Persistent multi-agent teams (F16, L).** Targets: extend `internal/agent` coordinator + `internal/roocode` with named-team persistence. AC: coordinator delegates to specialists with isolated context; team state survives restart (§11.4.119 single-resource-owner for shared files).
- **T4.3 — Async delegate / background sub-sessions (F45, M).** Targets: `internal/workflow/background.go` + `internal/roocode`. AC: background sub-session runs in parallel, summary injected into main context (§11.4.89 background execution).
- **T4.4 — Cross-AI peer review (F63, M).** Targets: `internal/verifier` ensemble → multi-runtime reviewer. AC: a second model/runtime reviews a diff; findings captured.
- **T4.5 — Kanban/parallel task board + MCP socket pool (F21, F66, L).** Targets: app/server task board; `internal/mcp` shared-process pool. AC: parallel cards each on own worktree; MCP procs shared with measured memory reduction.

Phase 5 — Connectors, Server & Headless Surface

- **T5.1 — Robust headless output (F19, M, gated on ST0.1.1).** Targets: `cmd/cli --output-format json|stream-json`. AC: NDJSON event stream; `git diff | helix --json` works in a pipe; `-count=3` deterministic (§11.4.98).
- **T5.2 — OpenAI-compatible server (F56, M, gated on ST0.1.4).** Targets: `internal/server` chat-completions/responses/embeddings + SSE. AC: a real OpenAI client hits the endpoint and gets a completion.
- **T5.3 — Shared-session daemon (F31, L).** Targets: `internal/server` multi-client shared session keyed by cwd (crush/qwen/shai pattern). AC: two clients mirror the same session live; loopback no-auth, remote token-gated.
- **T5.4 — ACP mode (F35, M).** Targets: `cmd/cli --acp` JSON-RPC over stdio. AC: a JSON-RPC initialize/prompt round-trips (Zed/JetBrains-compatible shape).

- **T5.5 — Messaging connectors (F18, L, extend Herald per §11.4.74).** Targets: connector adapters → session mapping with access control. AC: a real Slack/ Telegram thread drives a session with @-attribution (§11.4.104) + intent recognition (§11.4.105).
- **T5.6 — Chainable `--trace` (F55, S).** AC: `helix ... | helix "now run it"` pipes a real conversation trace.

Phase 6 — Spec-Driven & Quality-Gate Workflow

- **T6.1 — Spec-driven command surface (F61, L).** Targets: `cmd/cli` + `internal/planner/plantree` — `/specify`,`/clarify`,`/plan`,`/tasks`,`/implement`,`/analyze` with a project constitution file. AC: a spec produces a plan → tasks → real implementation; constitution governs phases. (Strong synergy with HelixCode's existing constitution discipline.)
- **T6.2 — Recipes (parametrized reusable workflows) (F46, M).** Targets: `internal/plugins`/skills — parametrized recipe primitive. AC: a recipe runs with substituted parameters end-to-end.
- **T6.3 — Event triggers (file-watch + cron → skill/recipe) (F59, F17, M).** Targets: new `internal/triggers` daemon (file-watch + cron) firing skills. AC: a file change fires a subscribed skill; a cron schedule survives restart.
- **T6.4 — Whole-project generation one-shot (F51, M).** Targets: `cmd/cli` scaffold mode reading a prompt file → full project. AC: `helix gen <dir>` from a prompt file produces a buildable project.

Phase 7 — Long-Tail / Niche Features

- **T7.1 — Watch-files / IDE-comment mode (F09, M).** AC: an `AI!` comment in a file triggers an edit.
- **T7.2 — Lessons system (auto-injected guidance) (F47, M).** AC: a keyword/ tool match injects a lesson; opt-out works.
- **T7.3 — Introspect tool + self-config skill (F44, F32, S each).** AC: "what can you do?" answered from bundled docs; "configure yourself" edits config.
- **T7.4 — Model steering mid-task (F38, M).** AC: a steer message redirects without restarting the loop.
- **T7.5 — Profiles (YAML behavior presets) (F50, S).** AC: `--profile` switches a named behavior set.
- **T7.6 — Copy-context to web chat (F10, S).** AC: `/copy-context` yields paste-ready context.
- **T7.7 — Benchmark harness (F52, L) + ACI windowed editor (F53, M).** AC: agent runs against a public benchmark task; windowed viewer shows ~100 lines/turn with in-file search.
- **T7.8 — Computer use / desktop control (F48, L).** AC: a GUI action is driven + verified (compose §11.4.107 liveness).
- **T7.9 — Stateful code-first execution (DataFrames persist) (F65, L)** + **token-economy review** — only if a data-analytics use case is greenlit.

---

9. Dependency Touch-Map

Which org submodules / internal subsystems each feature cluster touches (per §11.4.51 equal-codebase, extend-don't-reimplement):

Feature cluster	HelixAgent	LLMsVerifier	HelixMemory (cognee)	HelixLLM	HelixSpecifie
Modes/Autonomy (F04,F11,F23)	✓ agent loop	✓ model status	—	✓ routing	—
Checkpoints/Rewind/ Tangent/Branches (F12,F25,F36,F43)	✓ session	—	partial (history)	—	—
Diff-sandbox (F22)	✓ worktree	—	—	—	—
Multi-agent/Delegate (F16,F45,F63)	✓ coordinator/ roocode	✓ ensemble	✓ shared mem	✓	—
Server/Headless/ACP/ Connectors (F19,F31,F35,F56,F18)	✓	✓	—	✓	—
Spec-driven (F61) + Recipes/Triggers (F46,F59)	✓ planner/ plantree	—	—	—	✓ (spec engine)
Sandbox/Lint-fix/ Auto-debug (F33,F06,F24)	✓ worker/ editor	✓ verifier	—	—	—
Context/@-mention/ RAG (F26,F42)	✓ context	—	✓ cognee	—	—

`HelixSpecifier` is the natural owner for F61's spec engine (catalogue-check before greenfield). `Herald` is the catalogue-first base for F18 messaging connectors.

---

10. Honest Effort Caveats

- **No over-claiming "missing".** Many marquee features (repo-map F01, auto-commit F02, voice F05, model routing F37, worktrees F39, RAG/knowledge F42, condense F57, sessions F58, approval F29/F34, LSP F28, provider catalogue F30) are **already HAVE** in HelixCode — porting work there is zero or surface-only.
- **PARTIAL ≠ trivial.** F12/F22/F33/F61 have substrate but the user-facing surface + cross-platform + anti-bluff coverage are the real cost (each is M–L).
- **Vendor-doc stubs.** codex `docs/*.md` and mistral-code's inherited gemini docs link out to websites; their feature claims were taken from README/AGENTS/ crate-names, not in-tree prose — do not assume the in-repo copy is the spec.
- **Out-of-scope rows.** F60 (IDE extension), F70 (account-plan auth), F52 (agent benchmark) are flagged but are app-team / provider / research concerns, not core-CLI ports.
- **`openhands` absent.** The operator's example list named openhands; it is NOT in `cli_agents/` — excluded.
- This is a **planning document** (§11.4.73 spec-versioning applies once accepted). No code written. Effort estimates are S/M/L bands, not hour commitments.

---

11. Sources Verified

Researched 2026-06-14 by read-only subagents against the in-repo vendored copies under `cli_agents/`. Primary evidence files (non-exhaustive):

- aider: `aider/README.md`, `aider/aider/website/docs/{repomap,usage/modes,usage/commands,usage/voice,usage/watch,usage/lint-test,usage/copypaste,usage/images-urls}.md`
- cline: `cline/README.md`, `cline/.clinerules/*`
- plandex: `plandex/README.md`, `plandex/docs/docs/core-concepts/{plans,reviewing-changes,context-management,autonomy,execution-and-debugging,version-control,branches}.md`, `plandex/docs/docs/repl.md`
- crush: `crush/README.md`, `crush/docs/hooks/`
- codex: `codex/README.md`, `codex/AGENTS.md`, `codex/docs/{sandbox,exec,skills,slash_commands,config,agents_md,execpolicy,authentication}.md`, `codex/codex-rs/` crate layout
- gemini-cli: `gemini-cli/README.md`, `gemini-cli/docs/cli/{plan-mode,rewind,model-routing,model-steering,git-worktrees,acp-mode,skills,headless}.md`, `gemini-cli/docs/{checkpointing,hooks/index}.md`, `gemini-cli/docs/tools/*`
- qwen-code: `qwen-code/README.md`, `qwen-code/.qwen/skills/`
- copilot-cli: `copilot-cli/README.md`
- amazon-q: `amazon-q/docs/{built-in-tools,agent-format,knowledge-management,tangent-mode,introspect-tool,hooks,experiments,todo-lists}.md`
- mistral-code: `mistral-code/README.md`, `mistral-code/MISTRAL.md`, `mistral-code/docs/*` (mostly inherited)
- codename-goose: `codename-goose/README.md`, `ARCHITECTURE.md`, `crates/`, `documentation/blog/2025-02-21-gooseteam-mcp/index.md`
- gptme: `gptme/README.md`
- open-interpreter: `open-interpreter/README.md`
- forge: `forge/README.md`
- gpt-engineer: `gpt-engineer/README.md`
- swe-agent: `swe-agent/README.md`, `swe-agent/docs/background/aci.md`, `swe-agent/tools/`
- shai: `shai/README.md`
- nanocoder: `nanocoder/README.md`, `nanocoder/CLAUDE.md`, `nanocoder/docs/features/*`
- claude-squad: `claude-squad/README.md`
- claude-code: `claude-code/README.md`, `claude-code/plugins/`
- vtcode: `vtcode/README.md`
- warp: `warp/README.md`, `.warp/` skills
- get-shit-done: `get-shit-done/docs/FEATURES.md`, `commands/gsd/`, `AGENTS.md`
- spec-kit: `spec-kit/README.md`
- junie: `junie/README.md`
- aichat: `aichat/README.md`
- codai: `codai/README.md`
- taskweaver: `taskweaver/README.md`
- octogen: `octogen/README.md`
- zeroshot: `zeroshot/AGENTS.md`
- multiagent-coding: `multiagent-coding/README.md`
- agent-deck: `agent-deck/llms.txt`

- bridle: `bridle/CLAUDE.md`, `bridle/AGENTS.md`, `.claude-plugin/marketplace.json` (marketplace, not an agent)

HelixCode baseline verified against: `helix_code/internal/{agent,repomap,mcp,plugins,verifier,session,hooks,worker,memory,voice,clarification,task,autocommit,roocode,kilocode,planner,plantree,approval,approvalwire,quality,server,telemetry,context}` and `helix_code/applications/*`, plus `docs/plans/HXC-031-codex-cline-port.md`.

Sources verified 2026-06-22: cli_agents// *{README,AGENTS,CLAUDE}.md* , *helix_code/ internal/* , helix_code/applications/* , docs/plans/ HXC-031-codex-cline-port.md

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced repo trees, following the `docs/ARCHITECTURE.md` precedent — this is a port-mapping PLAN, not operator instructions for an external service). The plan's own `Method` field already declares it: read-only research of `cli_agents/<agent>/` READMEs+docs cross-checked against `helix_code/internal/*` + `helix_code/applications/*`. Cross-referenced on 2026-06-22: - The rev2 reconciliation paths it corrected (`internal/autonomy` → `internal/workflow/autonomy`, `internal/planner` → `internal/workflow/planmode`) are real moves in the live tree — the rev2 status update IS the §11.4.99 correction record for this doc. - **Honest status preserved:** the doc is explicitly `Status: DRAFT — research + plan only, NO code` and Phases 2–7 are marked unimplemented; no green bluff. - **`cli_agents/* = 50 directories`** present (the vendored reference catalogue the per-agent doc-paths point at). No external-service version claim → no upstream-staleness check applies (vendored reference copies, not a live service this doc instructs against).